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4.1.3. Student Information

01:49

Write a program to read a text file containing student information (name, age, and grade) using Pandas. Perform the following tasks:

- Display the first five rows of the data frame.
- Calculate the average age of the students (limit the average age up to 2 decimal places).
- Filter out the students who have a grade above a certain threshold (consider the threshold grade is '**B**').

Note:

Refer to the displayed test cases for better understanding.

```
import pandas as pd
```

```
# Read the text file into a DataFrame
file = input()
data = pd.read_csv(file, sep="\s+", header=None, names=["Name", "Age", "Grade"])
# Display the first five rows
print("First five rows:")
print(data.head())

# Calculate the average age (2 decimal places)
avg_age = round(data["Age"].mean(), 2)
print("Average age:", avg_age)

# Filter students with grade up to 'B' (A and B)
filtered = data[data["Grade"].isin(['A', 'B'])]

print("Students with a grade up to B")
```

print(filtered)

Average time
0.028 s
27.50 ms

Maximum time
0.040 s
40.00 ms

✓ 1 out of 1 shown test case(s) passed

✓ 1 out of 1 hidden test case(s) passed

✓ Test case 1 40 ms Debug

Expected output

Actual output

studentdata.txt

First five rows:

...Name...Age...Grade

0..John...25....A

1..Alin...22....B

2..Emma...24....A

3..Mike...23....C

4..Sony...21....B

Average age: 23.29

Students with a grade up to B

...Name...Age...Grade

0..John...25....A

1..Alin...22....B

2..Emma...24....A

4..Sony...21....B

5..Amia...26....A

studentdata.txt

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Terminal

Test cases

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